

Problem

A consumer has an annual consumption of 1200 MWh with a maximum demand of 2.4 MVA. The maximum demand charge is \$30 per kVA per annum, and the energy charge per kWh is 4 cents.

(a) Determine the annual cost of energy.

(b) Calculate the charge per kWh with a flat-rate tariff if the revenue to the utility company is to remain the same as for the two-part tariff.

Step-by-step solution

Step 1 of 2

(a)

Maximum demand charge = $2,400 \times 30$

Maximum demand charge = \$ 72,000

Energy cost = $\$ 0.04 \times 1,200 \times 10^3$

Energy cost = \$ 48, 00

Total charge = \$ 120,000.

Step 2 of 2

(b)

To obtain \$120,00 from 1,200 MWh will require a flat rate of,

$$= \frac{\$120,000}{1,200 \times 10^3} \text{ Per kWh}$$

$$= \$ 0.10 \text{ per kWh}$$